

| All Models Across All Datasets — Grouped by Algorithm |         |           |       |           |       |          |       |          |       |                 |       |        |       |                |       |        |       |                    |       |        |       |                   |       |        |       |
|-------------------------------------------------------|---------|-----------|-------|-----------|-------|----------|-------|----------|-------|-----------------|-------|--------|-------|----------------|-------|--------|-------|--------------------|-------|--------|-------|-------------------|-------|--------|-------|
| Model information                                     |         | Train acc |       | Valid acc |       | Test acc |       | T-H rate |       | Healthy (H) P&R |       |        |       | Glioma (G) P&R |       |        |       | Meningioma (M) P&R |       |        |       | Pituitary (P) P&R |       |        |       |
| Algorithm                                             | Dataset | Mean      | SD    | Mean      | SD    | Mean     | SD    | Mean     | SD    | P               | P SD  | R Mean | R SD  | P Mean         | P SD  | R Mean | R SD  | P Mean             | P SD  | R Mean | R SD  | P Mean            | P SD  | R Mean | R SD  |
| AlexNet                                               | dataB   | 0.979     | 0.002 | 0.952     | 0.004 | 0.946    | 0.003 | 0.0058   | 0.001 | 0.985           | 0.004 | 0.975  | 0.008 | 0.949          | 0.009 | 0.917  | 0.008 | 0.892              | 0.005 | 0.910  | 0.006 | 0.958             | 0.015 | 0.974  | 0.013 |
|                                                       | dataBC  | 0.970     | 0.019 | 0.879     | 0.021 | 0.870    | 0.016 | 0.0252   | 0.005 | 0.928           | 0.015 | 0.917  | 0.022 | 0.916          | 0.034 | 0.887  | 0.027 | 0.795              | 0.018 | 0.740  | 0.052 | 0.879             | 0.034 | 0.938  | 0.024 |
|                                                       | dataB2  | 0.959     | 0.025 | 0.774     | 0.072 | 0.745    | 0.043 | 0.0722   | 0.024 | 0.806           | 0.044 | 0.753  | 0.091 | 0.808          | 0.075 | 0.823  | 0.092 | 0.666              | 0.094 | 0.527  | 0.136 | 0.790             | 0.064 | 0.877  | 0.035 |
| VGG-16                                                | dataB   | 0.986     | 0.001 | 0.960     | 0.001 | 0.957    | 0.005 | 0.0079   | 0.004 | 0.980           | 0.011 | 0.979  | 0.014 | 0.968          | 0.012 | 0.940  | 0.008 | 0.915              | 0.009 | 0.926  | 0.003 | 0.962             | 0.010 | 0.975  | 0.004 |
|                                                       | dataBC  | 0.976     | 0.016 | 0.882     | 0.021 | 0.870    | 0.012 | 0.0189   | 0.007 | 0.945           | 0.018 | 0.906  | 0.023 | 0.899          | 0.023 | 0.904  | 0.029 | 0.780              | 0.033 | 0.756  | 0.040 | 0.890             | 0.021 | 0.913  | 0.042 |
|                                                       | dataB2  | 0.979     | 0.017 | 0.801     | 0.049 | 0.794    | 0.020 | 0.0611   | 0.024 | 0.851           | 0.051 | 0.823  | 0.042 | 0.850          | 0.040 | 0.897  | 0.069 | 0.725              | 0.081 | 0.597  | 0.117 | 0.836             | 0.066 | 0.860  | 0.056 |
| VGG-19                                                | dataB   | 0.985     | 0.007 | 0.957     | 0.004 | 0.957    | 0.007 | 0.0081   | 0.004 | 0.979           | 0.010 | 0.976  | 0.010 | 0.962          | 0.009 | 0.950  | 0.026 | 0.931              | 0.030 | 0.917  | 0.022 | 0.953             | 0.004 | 0.979  | 0.007 |
|                                                       | dataBC  | 0.968     | 0.012 | 0.885     | 0.025 | 0.865    | 0.026 | 0.0222   | 0.009 | 0.938           | 0.022 | 0.906  | 0.025 | 0.911          | 0.038 | 0.926  | 0.022 | 0.797              | 0.046 | 0.718  | 0.087 | 0.859             | 0.036 | 0.912  | 0.045 |
|                                                       | dataB2  | 0.947     | 0.027 | 0.785     | 0.061 | 0.743    | 0.031 | 0.0422   | 0.018 | 0.877           | 0.052 | 0.760  | 0.086 | 0.775          | 0.069 | 0.827  | 0.080 | 0.664              | 0.120 | 0.537  | 0.087 | 0.815             | 0.041 | 0.850  | 0.045 |
| EfficientNet                                          | dataB   | 0.983     | 0.001 | 0.946     | 0.001 | 0.944    | 0.002 | 0.0122   | 0.002 | 0.969           | 0.004 | 0.992  | 0.001 | 0.968          | 0.002 | 0.907  | 0.008 | 0.885              | 0.006 | 0.891  | 0.015 | 0.952             | 0.009 | 0.974  | 0.005 |
|                                                       | dataBC  | 0.989     | 0.003 | 0.868     | 0.015 | 0.871    | 0.012 | 0.0474   | 0.008 | 0.873           | 0.021 | 0.930  | 0.017 | 0.912          | 0.021 | 0.912  | 0.014 | 0.818              | 0.014 | 0.692  | 0.036 | 0.895             | 0.016 | 0.948  | 0.017 |
|                                                       | dataB2  | 0.758     | 0.028 | 0.626     | 0.054 | 0.620    | 0.038 | 0.2244   | 0.023 | 0.595           | 0.032 | 0.900  | 0.070 | 0.569          | 0.048 | 0.807  | 0.044 | 0.417              | 0.204 | 0.163  | 0.081 | 0.919             | 0.117 | 0.610  | 0.052 |
| LinThoUCB                                             | dataB   | 0.973     | 0.001 | 0.947     | 0.002 | 0.944    | 0.003 | 0.0149   | 0.001 | 0.963           | 0.003 | 0.988  | 0.003 | 0.954          | 0.005 | 0.919  | 0.005 | 0.895              | 0.004 | 0.887  | 0.010 | 0.959             | 0.003 | 0.970  | 0.004 |
|                                                       | dataBC  | 0.994     | 0.005 | 0.897     | 0.013 | 0.888    | 0.006 | 0.0452   | 0.003 | 0.878           | 0.008 | 0.953  | 0.008 | 0.913          | 0.010 | 0.922  | 0.024 | 0.855              | 0.018 | 0.726  | 0.035 | 0.920             | 0.012 | 0.951  | 0.016 |
|                                                       | dataB2  | 0.992     | 0.007 | 0.875     | 0.023 | 0.800    | 0.026 | 0.0935   | 0.016 | 0.769           | 0.020 | 0.863  | 0.040 | 0.821          | 0.043 | 0.866  | 0.051 | 0.723              | 0.052 | 0.554  | 0.078 | 0.908             | 0.034 | 0.916  | 0.049 |
| NthoUCB                                               | dataB   | 0.928     | 0.001 | 0.919     | 0.002 | 0.912    | 0.003 | 0.0125   | 0.001 | 0.968           | 0.004 | 0.970  | 0.004 | 0.919          | 0.014 | 0.861  | 0.014 | 0.827              | 0.013 | 0.847  | 0.007 | 0.931             | 0.005 | 0.953  | 0.007 |
|                                                       | dataBC  | 0.979     | 0.005 | 0.909     | 0.009 | 0.875    | 0.011 | 0.0207   | 0.010 | 0.940           | 0.026 | 0.927  | 0.023 | 0.891          | 0.012 | 0.878  | 0.031 | 0.787              | 0.031 | 0.756  | 0.044 | 0.900             | 0.016 | 0.940  | 0.018 |
|                                                       | dataB2  | 0.988     | 0.015 | 0.914     | 0.014 | 0.836    | 0.029 | 0.0413   | 0.008 | 0.884           | 0.016 | 0.780  | 0.013 | 0.880          | 0.044 | 0.879  | 0.043 | 0.780              | 0.074 | 0.743  | 0.042 | 0.878             | 0.036 | 0.948  | 0.053 |
| NEG                                                   | dataB   | 0.929     | 0.003 | 0.918     | 0.001 | 0.912    | 0.002 | 0.0188   | 0.002 | 0.953           | 0.005 | 0.975  | 0.003 | 0.924          | 0.013 | 0.872  | 0.011 | 0.835              | 0.011 | 0.832  | 0.015 | 0.932             | 0.007 | 0.954  | 0.011 |
|                                                       | dataBC  | 0.980     | 0.007 | 0.913     | 0.009 | 0.867    | 0.017 | 0.0267   | 0.005 | 0.922           | 0.015 | 0.907  | 0.018 | 0.896          | 0.015 | 0.896  | 0.026 | 0.768              | 0.029 | 0.751  | 0.040 | 0.903             | 0.032 | 0.916  | 0.024 |
|                                                       | dataB2  | 0.989     | 0.005 | 0.901     | 0.015 | 0.805    | 0.032 | 0.0543   | 0.010 | 0.851           | 0.025 | 0.803  | 0.058 | 0.839          | 0.015 | 0.900  | 0.038 | 0.677              | 0.082 | 0.551  | 0.108 | 0.856             | 0.078 | 0.962  | 0.031 |
| linEG                                                 | dataB   | 0.971     | 0.001 | 0.944     | 0.004 | 0.939    | 0.001 | 0.0168   | 0.001 | 0.958           | 0.003 | 0.987  | 0.002 | 0.943          | 0.003 | 0.917  | 0.003 | 0.892              | 0.005 | 0.877  | 0.005 | 0.956             | 0.004 | 0.962  | 0.003 |
|                                                       | dataBC  | 0.996     | 0.000 | 0.889     | 0.012 | 0.866    | 0.012 | 0.0526   | 0.009 | 0.859           | 0.023 | 0.931  | 0.019 | 0.889          | 0.012 | 0.900  | 0.020 | 0.826              | 0.022 | 0.718  | 0.033 | 0.903             | 0.033 | 0.916  | 0.025 |
|                                                       | dataB2  | 0.989     | 0.009 | 0.884     | 0.019 | 0.760    | 0.024 | 0.1109   | 0.045 | 0.731           | 0.078 | 0.832  | 0.084 | 0.804          | 0.069 | 0.844  | 0.036 | 0.648              | 0.045 | 0.509  | 0.087 | 0.872             | 0.055 | 0.857  | 0.049 |
| Boltz1.5                                              | dataB   | 0.923     | 0.001 | 0.913     | 0.002 | 0.906    | 0.001 | 0.0175   | 0.001 | 0.956           | 0.003 | 0.966  | 0.006 | 0.915          | 0.019 | 0.862  | 0.016 | 0.828              | 0.007 | 0.822  | 0.014 | 0.921             | 0.011 | 0.960  | 0.007 |
|                                                       | dataBC  | 0.984     | 0.009 | 0.917     | 0.005 | 0.872    | 0.013 | 0.0289   | 0.009 | 0.915           | 0.026 | 0.913  | 0.018 | 0.887          | 0.014 | 0.911  | 0.021 | 0.786              | 0.028 | 0.738  | 0.041 | 0.908             | 0.025 | 0.925  | 0.017 |
|                                                       | dataB2  | 0.984     | 0.018 | 0.907     | 0.022 | 0.825    | 0.012 | 0.0457   | 0.014 | 0.866           | 0.042 | 0.762  | 0.025 | 0.850          | 0.031 | 0.900  | 0.012 | 0.755              | 0.046 | 0.657  | 0.061 | 0.895             | 0.052 | 0.969  | 0.020 |
| Boltz2                                                | dataB   | 0.925     | 0.003 | 0.915     | 0.002 | 0.909    | 0.004 | 0.0188   | 0.001 | 0.953           | 0.003 | 0.973  | 0.004 | 0.923          | 0.019 | 0.862  | 0.010 | 0.834              | 0.007 | 0.825  | 0.019 | 0.919             | 0.002 | 0.957  | 0.007 |
|                                                       | dataBC  | 0.989     | 0.004 | 0.916     | 0.002 | 0.868    | 0.010 | 0.0237   | 0.006 | 0.930           | 0.017 | 0.911  | 0.012 | 0.895          | 0.016 | 0.887  | 0.036 | 0.785              | 0.020 | 0.741  | 0.033 | 0.879             | 0.020 | 0.936  | 0.026 |
|                                                       | dataB2  | 0.984     | 0.007 | 0.896     | 0.006 | 0.836    | 0.020 | 0.0630   | 0.013 | 0.834           | 0.016 | 0.839  | 0.054 | 0.854          | 0.050 | 0.914  | 0.029 | 0.757              | 0.052 | 0.680  | 0.039 | 0.942             | 0.040 | 0.910  | 0.044 |
| linBoltz1.5                                           | dataB   | 0.959     | 0.001 | 0.941     | 0.002 | 0.937    | 0.001 | 0.0181   | 0.002 | 0.955           | 0.004 | 0.989  | 0.001 | 0.953          | 0.004 | 0.906  | 0.006 | 0.885              | 0.005 | 0.873  | 0.006 | 0.951             | 0.004 | 0.968  | 0.002 |
|                                                       | dataBC  | 0.992     | 0.004 | 0.894     | 0.013 | 0.889    | 0.006 | 0.0437   | 0.009 | 0.884           | 0.023 | 0.960  | 0.015 | 0.918          | 0.010 | 0.931  | 0.020 | 0.853              | 0.019 | 0.724  | 0.028 | 0.916             | 0.022 | 0.943  | 0.023 |
|                                                       | dataB2  | 0.996     | 0.003 | 0.868     | 0.020 | 0.810    | 0.030 | 0.1022   | 0.021 | 0.765           | 0.031 | 0.884  | 0.044 | 0.839          | 0.017 | 0.868  | 0.053 | 0.740              | 0.058 | 0.568  | 0.077 | 0.921             | 0.047 | 0.916  | 0.034 |
| linBoltz2                                             | dataB   | 0.959     | 0.001 | 0.941     | 0.002 | 0.937    | 0.001 | 0.0181   | 0.002 | 0.955           | 0.004 | 0.989  | 0.001 | 0.953          | 0.004 | 0.906  | 0.006 | 0.885              | 0.005 | 0.873  | 0.006 | 0.951             | 0.004 | 0.968  | 0.002 |
|                                                       | dataBC  | 0.992     | 0.004 | 0.894     | 0.013 | 0.889    | 0.006 | 0.0437   | 0.009 | 0.884           | 0.023 | 0.960  | 0.015 | 0.918          | 0.010 | 0.931  | 0.020 | 0.853              | 0.019 | 0.724  | 0.028 | 0.916             | 0.022 | 0.943  | 0.023 |
|                                                       | dataB2  | 0.996     | 0.003 | 0.868     | 0.020 | 0.810    | 0.030 | 0.1022   | 0.021 | 0.765           | 0.031 | 0.884  | 0.044 | 0.839          | 0.017 | 0.868  | 0.053 | 0.740              | 0.058 | 0.568  | 0.077 | 0.921             | 0.047 | 0.916  | 0.034 |